

Polynomial factoring

Question 1

Write the polynomial $X^3 + 7X + 7$ as a product of irreducible factors in $\mathbb{Q}[X]$.

Solution. This is a cubic polynomial. Hence, it is reducible if and only if it has a root.

This is a monic polynomial in $\mathbb{Z}[X]$. Hence, if it is reducible then the factors can be taken as monic polynomials in $\mathbb{Z}[X]$.

Since 7 is prime, its only factors are ± 1 and ± 7 .

Thus, the only possible monic linear factors are $X \pm 1$ and $X \pm 7$.

We check

$$\begin{aligned}1^3 + 7 \cdot 1 + 7 &= 15 \\ (-1)^3 + 7(-1) + 7 &= -1 \\ 7^3 + 7 \cdot 7 + 7 &= 399 \\ (-7)^3 + 7(-7) + 7 &= -385\end{aligned}$$

Thus, the polynomial is irreducible in $\mathbb{Q}[X]$.

Question 2

Write the polynomial $X^3 + 7X + 7$ as a product of irreducible factors in $\mathbb{F}_3[X]$.

Solution. We note that $1^3 + 7 \cdot 1 + 7 = X^3 + X + 1 \pmod{3}$.

This is a cubic polynomial. Hence, it is reducible if and only if it has a root.

We check $1^3 + 1 \cdot 1 + 1 = 0 \pmod{3}$.

Dividing by $X - 1$ we get $X^3 + X + 1 = (X - 1)(X^2 + X - 1)$.

The polynomial $X^2 + X - 1$ is of degree 2. Hence, it is reducible if and only if it has a root.

We check

$$\begin{aligned}1^2 + 1 \cdot 1 - 1 &= 1 \pmod{3} \\ (-1)^2 + 1 \cdot (-1) - 1 &= -1 \pmod{3} \\ 0^2 + 1 \cdot 0 - 1 &= -1 \pmod{3}\end{aligned}$$

Thus, the polynomial $X^2 + X - 1$ is irreducible in $\mathbb{F}_3[X]$.

The required factoring is $X^3 + 7X + 7 = (X - 1)(X^2 + X - 1)$ in $\mathbb{F}_3[X]$.